

Lecomte Antoine

2) a) b)

c) d)

```
upstream lerebours{
    server monsite1.backend:8080;
    server monsite2.backend:8081;
}

server {
    listen 443 ssl;
    server_name monsite.lerebours.fr;

    #include /etc/nginx/security/restrict;
    add_header Organization 'lerebours';
    ssl_certificate /etc/nginx/ssl/nginx-selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/nginx-selfsigned.key;

    location / {
        proxy_pass http://upstream_lerebours;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

e)

```
yohan@debian:/etc/nginx/sites-available$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

f) configuration file /etc/nginx/nginx.conf :

```
user www-data; worker_processes auto; worker_cpu_affinity auto; pid /run/nginx.pid;
error_log /var/log/nginx/error.log; include /etc/nginx/modules-enabled/*.conf;
```

```
events { worker_connections 768; # multi_accept on; }
```

```
http {
```

```
##
```

```
# Basic Settings
```

```
##
```

```
sendfile on;
```

```
tcp_nopush on;
```

```
types_hash_max_size 2048;
```

```
server_tokens off; # Recommended practice is to turn this off
```

```
# server_names_hash_bucket_size 64;
```

```
# server_name_in_redirect off;
```

```
include /etc/nginx/mime.types;
```

```
default_type application/octet-stream;
```

```
##
```

```
# SSL Settings
```

```
##
```

```
ssl_protocols TLSv1.2 TLSv1.3; # Dropping SSLv3 (POODLE), TLS 1.0, 1.1
```

```
ssl_prefer_server_ciphers off; # Don't force server cipher order.
```

```
##
```

```
# Logging Settings
```

```
##
```

```
access_log /var/log/nginx/access.log;
```

```
##
```

```
# Gzip Settings
```

```
##
```

```
gzip on;
```

```
# gzip_vary on;
```

```
# gzip_proxied any;
```

```
# gzip_comp_level 6;
```

```
# gzip_buffers 16 8k;
```

```
# gzip_http_version 1.1;
```

```
# gzip_types text/plain text/css application/json application/javascript text/xml  
application/xml application/xml+rss text/javascript;
```

```
##
```

```
# Virtual Host Configs
```

```
##
```

```
include /etc/nginx/conf.d/*.conf;
```

```
include /etc/nginx/sites-enabled/*;
```

```
}
```

```
#mail {
```

See sample authentication script at:

<http://wiki.nginx.org/ImapAuthenticateWithApachePhpScript>

auth_http localhost/auth.php;

pop3_capabilities "TOP" "USER";

imap_capabilities "IMAP4rev1" "UIDPLUS";

server {

listen localhost:110;

protocol pop3;

proxy on;

}

server {

listen localhost:143;

protocol imap;

proxy on;

}

configuration file /etc/nginx/mime.types :

types { text/html html htm shtml; text/css css; text/xml xml; image/gif gif; image/jpeg
jpeg jpg; application/javascript js; application/atom+xml atom; application/rss+xml rss;

text/mathml mml;

text/plain txt;

text/vnd.sun.j2me.app-descriptor jad;

text/vnd.wap.wml wml;

text/x-component htc;

image/avif	avif;
image/png	png;
image/svg+xml	svg svgz;
image/tiff	tif tiff;
image/vnd.wap.wbmp	wbmp;
image/webp	webp;
image/x-icon	ico;
image/x-jng	jng;
image/x-ms-bmp	bmp;
font/woff	woff;
font/woff2	woff2;
application/java-archive	jar war ear;
application/json	json;
application/mac-binhex40	hqx;
application/msword	doc;
application/pdf	pdf;
application/postscript	ps eps ai;
application/rtf	rtf;
application/vnd.apple.mpegurl	m3u8;
application/vnd.google-earth.kml+xml	kml;
application/vnd.google-earth.kmz	kmz;
application/vnd.ms-excel	xls;
application/vnd.ms-fontobject	eot;
application/vnd.ms-powerpoint	ppt;
application/vnd.oasis.opendocument.graphics	odg;
application/vnd.oasis.opendocument.presentation	odp;
application/vnd.oasis.opendocument.spreadsheet	ods;
application/vnd.oasis.opendocument.text	odt;
application/vnd.openxmlformats-officedocument.presentationml.presentation	pptx;
application/vnd.openxmlformats-officedocument.spreadsheetml.sheet	xlsx;
application/vnd.openxmlformats-officedocument.wordprocessingml.document	docx;
application/vnd.wap.wmlc	wmlc;
application/wasm	wasm;
application/x-7z-compressed	7z;

application/x-cocoa	cco;
application/x-java-archive-diff	jardiff;
application/x-java-jnlp-file	jnlp;
application/x-makeself	run;
application/x-perl	pl pm;
application/x-pilot	prc pdb;
application/x-rar-compressed	rar;
application/x-redhat-package-manager	rpm;
application/x-sea	sea;
application/x-shockwave-flash	swf;
application/x-stuffit	sit;
application/x-tcl	tcl tk;
application/x-x509-ca-cert	der pem crt;
application/x-xpinstall	xpi;
application/xhtml+xml	xhtml;
application/xslt+xml	xsl xslt;
application/xspf+xml	xspf;
application/zip	zip;
application/octet-stream	bin exe dll;
application/octet-stream	deb;
application/octet-stream	dmg;
application/octet-stream	iso img;
application/octet-stream	msi msp msm;
audio/midi	mid midi kar;
audio/mpeg	mp3;
audio/ogg	ogg;
audio/x-m4a	m4a;
audio/x-realaudio	ra;
video/3gpp	3gpp 3gp;
video/mp2t	ts;
video/mp4	mp4;
video/mpeg	mpeg mpg;
video/ogg	ogv;
video/quicktime	mov;
video/webm	webm;
video/x-flv	flv;
video/x-m4v	m4v;
video/x-matroska	mkv;
video/x-mng	mng;

```
video/x-ms-asf          asx asf;
video/x-ms-wmv          wmv;
video/x-msvideo         avi;

}
```

configuration file /etc/nginx/conf.d/loadbalancer.conf :

```
upstream upstream_lerebours{ server monsite1.backend:8080; server
monsite2.backend:8081; }
```

```
server{ listen 443 ssl; server_name monsite.lerebours.fr;
```

```
    #include /etc/nginx/security/restrict;
    add_header Organization 'lerebours';
    ssl_certificate /etc/nginx/ssl/nginx-selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/nginx-selfsigned.key;
```

```
    location / {
        proxy_pass http://upstream_lerebours;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
```

```
}
```

configuration file /etc/nginx/sites-enabled/default :

You should look at the following URL's in order to grasp a solid understanding of Nginx configuration files in order to fully unleash the power of Nginx.

<https://www.nginx.com/resources/wiki/start/>

https://www.nginx.com/resources/wiki/start/topics/tutorials/config_pitfalls/

<https://wiki.debian.org/Nginx/DirectoryStructure>

In most cases, administrators will remove this file from sites-enabled/ and leave it as reference inside of sites-available where it will continue to be updated by the nginx packaging team.

This file will automatically load configuration files provided by other applications, such as Drupal or Wordpress. These applications will be made available underneath a path with that package name, such as /drupal8.

Please see /usr/share/doc/nginx-doc/examples/ for more detailed examples.

Default server configuration

```
server { listen 80 default_server; listen [::]:80 default_server;

# SSL configuration
#
# listen 443 ssl default_server;
# listen [::]:443 ssl default_server;
#
# Note: You should disable gzip for SSL traffic.
# See: https://bugs.debian.org/773332
#
# Read up on ssl_ciphers to ensure a secure configuration.
# See: https://bugs.debian.org/765782
```



```
#
# Self signed certs generated by the ssl-cert package
# Don't use them in a production server!
#
# include snippets/snakeoil.conf;

root /var/www/html;

# Add index.php to the list if you are using PHP
index index.html index.htm index.nginx-debian.html;

server_name _;

location / {
    # First attempt to serve request as file, then
    # as directory, then fall back to displaying a 404.
    try_files $uri $uri/ =404;
}

# pass PHP scripts to FastCGI server
#
#location ~ /\.php$ {
#    include snippets/fastcgi-php.conf;
#
#    # With php-fpm (or other unix sockets):
#    fastcgi_pass unix:/run/php/php7.4-fpm.sock;
#    # With php-cgi (or other tcp sockets):
#    fastcgi_pass 127.0.0.1:9000;
#}

# deny access to .htaccess files, if Apache's document root
# concurs with nginx's one
#
#location ~ /\.ht {
#    deny all;
#}

}
```

Virtual Host configuration for example.com

You can move that to a different file under sites-available/ and symlink that to sites-enabled/ to enable it.

```
#server {  
  
listen 80;  
  
listen [::]:80;  
  
server_name example.com;  
  
root /var/www/example.com;  
  
index index.html;  
  
location / {  
  
try_files $uri $uri/ =404;  
  
}  
  
#}
```

configuration file /etc/nginx/sites-enabled/sites:

```
server { listen 8080; server_name monsite1.backend; root /var/www/monsite1; index  
index.html; }  
  
server { listen 8081; server_name monsite2.backend; root /var/www/monsite2; index  
index.html; }
```

Le fichier loadbalancer.conf est bien pris en compte car on l'a rajouté dans /etc/nginx et qu'on a restart avec la commande.

g)

A screenshot of a terminal window showing the /etc/hosts file being edited with GNU nano 8.4. The file contains several entries for localhost and specific IP addresses pointing to various domains. A comment line indicates that the following lines are desirable for IPv6 capable hosts.

```
GNU nano 8.4 /etc/hosts
127.0.0.1 localhost
127.0.1.1 debian.myguest.virtualbox.org  debian
127.0.0.1 monsite1.backend
127.0.0.1 monsite2.backend
127.0.0.1 monsite.lerebours.fr
# The following lines are desirable for IPv6 capable hosts
::1      localhost ip6-localhost ip6-loopback
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

h) Une exception de sécurité est levée (Warning).

i) Lorsqu'on accède à <https://monsite.lerebours.fr>, Firefox établit bien une connexion HTTPS, mais il ne fait pas confiance au certificat SSL présenté par Nginx, et le certificat que l'on a généré étant signé par lui-même, et non par une autorité de certification reconnue, résumant pourquoi une exception de sécurité est levée, car l'authenticité du certificat ne peut être vérifiée.

j) Pour voir les caractéristiques du certificat SSL il suffit de se rendre dans les options avancées et d'afficher le certificat, puis détails. On peut voir que la date d'expiration est de 1 an exactement donc le 9 décembre 2026, taille de la clé publique : 2048 bits, le sujet est Le Rebours Paris XIII et même chose pour l'autorité de certificat qui l'a signé (provoque le warning).

Subject Name

Country FR
State/Province Ile de France
Locality PARIS
Organization Le Rebours Paris XIII
Organizational Unit CYBER
Common Name monsite.lerebours.fr
Email Address leconteyohan15@gmail.com

Issuer Name

Country FR
State/Province Ile de France
Locality PARIS
Organization Le Rebours Paris XIII
Organizational Unit CYBER
Common Name monsite.lerebours.fr
Email Address leconteyohan15@gmail.com

Validity

Not Before Tue, 09 Dec 2025 10:03:07 GMT
Not After Wed, 09 Dec 2026 10:03:07 GMT

Public Key Info

Algorithm RSA
Key Size 2048
Exponent 65537
Modulus CB:C9:24:45:FC:BA:4E:4E:5B:E7:D0:65:B2:AD:92:94:4D:46:43:57:B5:20:64:E...

Miscellaneous

Miscellaneous	
Serial Number	49:0C:01:A8:DC:AD:06:D4:55:7C:94:B0:4B:6B:4C:11:5C:E6:57:A4
Signature Algorithm	SHA-256 with RSA Encryption
Version	3
Download	PEM (cert) PEM (chain)
Fingerprints	
SHA-256	5C:ED:38:58:23:E3:30:F8:78:A1:1D:B5:2C:04:BC:7F:99:B5:CE:3F:51:DD:32:B4...
SHA-1	B4:5A:52:55:5B:C5:6A:94:E8:FC:CA:F5:75:51:69:11:F6:5F:AF:6D
Basic Constraints	
Certificate Authority	Yes
Subject Key ID	
Key ID	F7:5B:B4:B4:1E:81:E7:E0:E6:09:AE:DA:75:5E:BF:9D:39:FF:52:2E
Authority Key ID	
Key ID	F7:5B:B4:B4:1E:81:E7:E0:E6:09:AE:DA:75:5E:BF:9D:39:FF:52:2E

3)b)

```
GNU nano 8.4 /etc/nginx/conf.d/loadbalancer.conf *
upstream upstream_lerebours{
    server monsite1.backend:8080 weight=3;
    server monsite2.backend:8081;
}

server {
    listen 443 ssl;
    server_name monsite.lerebours.fr;

    #include /etc/nginx/security/restrict;
    add_header Organization 'lerebours';
    ssl_certificate /etc/nginx/ssl/nginx-selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/nginx-selfsigned.key;

    location / {
        proxy_pass http://upstream_lerebours;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

La répartition de la charge ici n'est plus équitable, (initialement en round-robin, où chaque serveur reçoit le même nombre de requêtes, 50% vers monsite1.backend / 50% vers monsite2.backend), on est ici avec pondération. On a 75% des requêtes vers monsite1 et 25% vers monsite2. C'est utile si l'objectif est de favoriser un serveur plus puissant, ou simuler une architecture hétérogène.

c)

```
GNU nano 8.4 /etc/nginx/conf.d/loadbalancer.conf
upstream upstream_lerebours{
    least_conn;
    server monsite1.backend:8080;
    server monsite2.backend:8081;
}

server {
    listen 443 ssl;
    server_name monsite.lerebours.fr;

    #include /etc/nginx/security/restrict;
    add_header Organization 'lerebours';
    ssl_certificate /etc/nginx/ssl/nginx-selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/nginx-selfsigned.key;

    location / {
        proxy_pass http://upstream_lerebours;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

d) Nginx envoie chaque nouvelle requête au serveur qui a le moins de connexions actives.

e)

```
GNU nano 8.4 /etc/nginx/conf.d/loadbalancer.conf
upstream upstream_lerebours{
    ip_hash;
    server monsite1.backend:8080;
    server monsite2.backend:8081;
}

server {
    listen 443 ssl;
    server_name monsite.lerebours.fr;

    #include /etc/nginx/security/restrict;
    add_header Organization 'lerebours';
    ssl_certificate /etc/nginx/ssl/nginx-selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/nginx-selfsigned.key;

    location / {
        proxy_pass http://upstream_lerebours;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

f) Il y a maintenant une persistance de session basée sur l'adresse IP du client. C'est le même backend qui reçoit toujours l'IP du client. Ici : monsite1.backend est toujours le serveur affiché après rafraîchissement de la page.

4) a)

```
GNU nano 8.4 /etc/nginx/conf.d/loadbalancer.conf *
upstream upstream_lerebours{
    server monsite1.backend:8080;
    server monsite2.backend:8081;
}

server {
    listen 80;
    server_name monsite.lerebours.fr;
    return 302 https://$host$request_uri;
    #return 301 https://$host$request_uri;

    #include /etc/nginx/security/restrict;
    add_header Organization 'lerebours';
    ssl_certificate /etc/nginx/ssl/nginx-selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/nginx-selfsigned.key;

    location / {
        proxy_pass http://upstream_lerebours;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

c) Code HTTP 302 au moment de la redirection HTTP → HTTPS, puis code HTTP 200 (accès au contenu avec Nginx) et code HTTP 200 (monsite1 et monsite2 en round-robin).

d) Redirection temporaire si on laisse return 302 ou redirection permanente avec 301.

e) La redirection permanente est adaptée ici (301) pour rediriger tout le trafic HTTP vers HTTPS de manière définitive pour le site.

5) b)

```
GNU nano 8.4 /etc/nginx/security/restrict *
##AuthBasic & Filtrage IP
allow 1.2.3.4;
deny all;
auth_basic "Access restricted";
auth_basic_user_file /etc/nginx/security/.htpasswd;
```

c)

```
yohan@debian:/etc/nginx$ sudo apt install apache2-utils
The following package was automatically installed and is no longer required:
  linux-image-6.12.43+deb13-amd64
Use 'sudo apt autoremove' to remove it.
```

```
Installing:
  apache2-utils
```

```
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 215 kB
  Space needed: 455 kB / 12.1 GB available
```

```
Get:1 http://deb.debian.org/debian trixie/main amd64 apache2-utils amd64 2.4.65-2 [215 kB]
Fetched 215 kB in 0s (576 kB/s)
Selecting previously unselected package apache2-utils.
(Reading database ... 146257 files and directories currently installed.)
Preparing to unpack .../apache2-utils_2.4.65-2_amd64.deb ...
Unpacking apache2-utils (2.4.65-2) ...
Setting up apache2-utils (2.4.65-2) ...
Processing triggers for man-db (2.13.1-1) ...
yohan@debian:/etc/nginx$ htpasswd -c /etc/nginx/security/.htpasswd lerebours
htpasswd: cannot create file /etc/nginx/security/.htpasswd
yohan@debian:/etc/nginx$ sudo htpasswd -c /etc/nginx/security/.htpasswd lerebours
New password:
Re-type new password:
Adding password for user lerebours
```

e) Ce n'est plus possible de se connecter au site.

f) allow <notre_IP>;

```
GNU nano 8.4 /etc/nginx/security/restrict
##AuthBasic & Filtrage IP
allow 1.2.3.4;
allow 193.252.176.167 ;
deny all;
auth_basic "Access restricted";
auth_basic_user_file /etc/nginx/security/.htpasswd;
```

6) a) Un seul point central gère les certificats SSL/TLS, donc meilleure maintenance pour renouveler les certificats et mêmes règles TLS pour tous les services (sécurité homogène). Les serveurs applicatifs n'ont pas besoin de gérer HTTPS, les backends communiquent en HTTP (performant).

b) Nginx est léger et performant et supporte nativement le reverse proxy et le load balancing. Il a une configuration assez simple et supporte SSL/TLS, HTTP, les headers et redirections, et est très fiable car il est utilisé énormément en production.

c) Round-Robin (défaut) : simple, équitable, mais ignore la charge réelle des serveurs.

Weight : s'adapte aux serveurs hétérogènes (certains doivent être prioritaires s'ils sont plus puissants) mais la charge est statique.

Least_conn : prend en compte la charge pour tenir compte des performances des serveurs mais ne tient pas compte des clients qui se sont déjà connectés pour leurs prochaines connexions.

IP_hash : permet à l'inverse la persistance de session, mais peut déséquilibrer la charge entre les serveurs.

Le choix adéquat dépend du type d'application : stateful / stateless.

d) On peut éviter l'exception de sécurité due au certificat en utilisant un certificat signé par une autorité reconnue (comme Let's Encrypt) ou en ajoutant le certificat auto-signé à la liste de confiance du navigateur, en vérifiant que le CN / SAN correspond bien au nom de domaine et que la chaîne de certificats est complète.

e) proxy_set_header permet de transmettre ou modifier les en-têtes HTTP envoyés au backend. Nous l'avons utilisé sous :

proxy_set_header Host \$host; → le backend peut connaître le nom de domaine original.

proxy_set_header X-Real-IP \$remote_addr; → transmet l'IP réelle du client.

Seule l'IP du proxy est visible par le backend sans cela.

f) add_header permet d'ajouter des en-têtes HTTP à la réponse envoyée au client.

Exemple : add_header Organization 'lerebours';

Ils peuvent servir pour la sécurité, le débogage ou pour l'envoi d'informations personnalisées dans la réponse.

g) L'authentification basique transmet seulement login et mot de passe en Base64. Ce n'est pas du chiffrement, c'est décodable, mais HTTPS chiffre totalement la communication, donc les identifiants ne peuvent pas être interceptés s'ils sont inconnus, c'est sécurisé.